



MATH120: Antiderivatives and Integrals

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Learning Objectives

- Calculate the mean, median, mode, and range of a data set
- Compute sample variance and standard deviation
- Construct quartiles, the IQR, and identify outliers
- Describe the shape, center, and spread of a distribution

Simplify each expression completely. Show all steps and circle your final answer.

Antiderivatives

1. Find the antiderivative (indefinite integral): integral of $6x^1 dx$.

$$\int 6x^1 dx$$

Answer: _____

2. The marginal cost function is $MC(x) = 2x^2$. Find the total cost function $TC(x)$ by finding the antiderivative. (Leave + C for the constant.)

$$\int 2x^2 dx$$

Answer: _____

3. Find the antiderivative (indefinite integral): integral of $1x^4 dx$.

$$\int 1x^4 dx$$

Answer: _____

4. The marginal cost function is $MC(x) = 4x^1$. Find the total cost function $TC(x)$ by finding the antiderivative. (Leave + C for the constant.)

$$\int 4x^1 dx$$

Answer: _____

5. Find the antiderivative (indefinite integral): integral of $4x^3 dx$.

$$\int 4x^3 dx$$

Answer: _____



6. The marginal cost function is $MC(x) = 3x^3$. Find the total cost function $TC(x)$ by finding the antiderivative. (Leave + C for the constant.)

$$\int 3x^3 dx$$

Answer: _____

7. Find the antiderivative (indefinite integral): integral of $2x^4 dx$.

$$\int 2x^4 dx$$

Answer: _____

8. The marginal cost function is $MC(x) = 2x^1$. Find the total cost function $TC(x)$ by finding the antiderivative. (Leave + C for the constant.)

$$\int 2x^1 dx$$

Answer: _____

9. Find the antiderivative (indefinite integral): integral of $5x^2 dx$.

$$\int 5x^2 dx$$

Answer: _____

10. The marginal cost function is $MC(x) = 3x^1$. Find the total cost function $TC(x)$ by finding the antiderivative. (Leave + C for the constant.)

$$\int 3x^1 dx$$

Answer: _____

11. Find the antiderivative (indefinite integral): integral of $5x^2 dx$.

$$\int 5x^2 dx$$

Answer: _____

12. The marginal cost function is $MC(x) = 5x^2$. Find the total cost function $TC(x)$ by finding the antiderivative. (Leave + C for the constant.)

$$\int 5x^2 dx$$

Answer: _____

13. Find the antiderivative (indefinite integral): integral of $1x^3 dx$.

$$\int 1x^3 dx$$

Answer: _____



14. The marginal cost function is $MC(x) = 2x^1$. Find the total cost function $TC(x)$ by finding the antiderivative. (Leave + C for the constant.)

$$\int 2x^1 dx$$

Answer: _____

15. Find the antiderivative (indefinite integral): integral of $1x^3 dx$.

$$\int 1x^3 dx$$

Answer: _____

16. The marginal cost function is $MC(x) = 4x^3$. Find the total cost function $TC(x)$ by finding the antiderivative. (Leave + C for the constant.)

$$\int 4x^3 dx$$

Answer: _____

Definite integrals

17. Evaluate the definite integral: integral from 2 to 2 of $x^2 dx$.

$$\int_2^2 x^2 dx$$

Answer: _____

18. Find the area under the curve $f(x) = x^2$ from $x = 1$ to $x = 3$.

$$\int_1^3 x^2 dx$$

Answer: _____

19. Evaluate the definite integral: integral from 1 to 3 of $x^3 dx$.

$$\int_1^3 x^3 dx$$

Answer: _____

20. Find the area under the curve $f(x) = x^1$ from $x = 1$ to $x = 3$.

$$\int_1^3 x^1 dx$$

Answer: _____



21. Evaluate the definite integral: integral from 0 to 5 of x^2 dx.

$$\int_0^5 x^2 dx$$

Answer: _____

22. Find the area under the curve $f(x) = x^1$ from $x = 1$ to $x = 4$.

$$\int_1^4 x^1 dx$$

Answer: _____

23. Evaluate the definite integral: integral from 0 to 5 of x^3 dx.

$$\int_0^5 x^3 dx$$

Answer: _____

24. Find the area under the curve $f(x) = x^1$ from $x = 1$ to $x = 4$.

$$\int_1^4 x^1 dx$$

Answer: _____

25. Evaluate the definite integral: integral from 1 to 3 of x^2 dx.

$$\int_1^3 x^2 dx$$

Answer: _____

26. Find the area under the curve $f(x) = x^2$ from $x = 0$ to $x = 4$.

$$\int_0^4 x^2 dx$$

Answer: _____

27. Evaluate the definite integral: integral from 0 to 5 of x^2 dx.

$$\int_0^5 x^2 dx$$

Answer: _____

28. Find the area under the curve $f(x) = x^2$ from $x = 1$ to $x = 3$.

$$\int_1^3 x^2 dx$$

Answer: _____



29. Evaluate the definite integral: integral from 1 to 2 of $x^3 dx$.

$$\int_1^2 x^3 dx$$

Answer: _____

30. Find the area under the curve $f(x) = x^1$ from $x = 0$ to $x = 2$.

$$\int_0^2 x^1 dx$$

Answer: _____





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ANSWER KEY & SOLUTIONS

Topics: Antiderivatives, Definite integrals. All answers verified by independent computation.

Solutions



Antiderivatives

1. Find the antiderivative (indefinite integral): integral of $6x^1 dx$.

$$\int 6x^1 dx$$

→ Power rule for integration: integral of $x^n dx = x^{(n+1)}/(n+1) + C$.

→ integral of $\{a\}x^{\{n\}} dx = (\{a\}/\{n+1\})x^{\{new_exp210\}} + C = \{answer_antideriv210\}x^{\{new_exp210\}} + C$.

Answer: $\frac{6}{2}x^2 + C \Rightarrow \text{coeff} = 3$

2. The marginal cost function is $MC(x) = 2x^2$. Find the total cost function $TC(x)$ by finding the antiderivative. (Leave + C for the constant.)

$$\int 2x^2 dx$$

→ $TC(x) = \text{integral of } MC(x) dx = \text{integral of } 2x^2 dx$.

→ $= 2/3x^3 + C$.

Answer: $\frac{2}{3}x^3 + C \Rightarrow \text{coeff} = 2/3$

3. Find the antiderivative (indefinite integral): integral of $1x^4 dx$.

$$\int 1x^4 dx$$

→ Power rule for integration: integral of $x^n dx = x^{(n+1)}/(n+1) + C$.

→ integral of $\{a\}x^{\{n\}} dx = (\{a\}/\{n+1\})x^{\{new_exp210\}} + C = \{answer_antideriv210\}x^{\{new_exp210\}} + C$.

Answer: $\frac{1}{5}x^5 + C \Rightarrow \text{coeff} = 1/5$

4. The marginal cost function is $MC(x) = 4x^1$. Find the total cost function $TC(x)$ by finding the antiderivative. (Leave + C for the constant.)

$$\int 4x^1 dx$$

→ $TC(x) = \text{integral of } MC(x) dx = \text{integral of } 4x^1 dx$.

→ $= 2x^2 + C$.

Answer: $\frac{4}{2}x^2 + C \Rightarrow \text{coeff} = 2$

5. Find the antiderivative (indefinite integral): integral of $4x^3 dx$.

$$\int 4x^3 dx$$

→ Power rule for integration: integral of $x^n dx = x^{(n+1)}/(n+1) + C$.

→ integral of $\{a\}x^{\{n\}} dx = (\{a\}/\{n+1\})x^{\{new_exp210\}} + C = \{answer_antideriv210\}x^{\{new_exp210\}} + C$.

Answer: $\frac{4}{4}x^4 + C \Rightarrow \text{coeff} = 1$



6. The marginal cost function is $MC(x) = 3x^3$. Find the total cost function $TC(x)$ by finding the antiderivative.
(Leave + C for the constant.)

$$\int 3x^3 dx$$

→ $TC(x) = \text{integral of } MC(x) dx = \text{integral of } 3x^3 dx.$

→ $= 3/4x^4 + C.$

Answer: $\frac{3}{4}x^4 + C \Rightarrow \text{coeff} = 3/4$

7. Find the antiderivative (indefinite integral): integral of $2x^4 dx$.

$$\int 2x^4 dx$$

→ Power rule for integration: $\text{integral of } x^n dx = x^{(n+1)}/(n+1) + C.$

→ $\text{integral of } \{a\}x^{\{n\}} dx = (\{a\}/\{n+1\})x^{\text{new_exp210}} + C = \{\text{answer_antideriv210}\}x^{\text{new_exp210}} + C.$

Answer: $\frac{2}{5}x^5 + C \Rightarrow \text{coeff} = 2/5$

8. The marginal cost function is $MC(x) = 2x^1$. Find the total cost function $TC(x)$ by finding the antiderivative.
(Leave + C for the constant.)

$$\int 2x^1 dx$$

→ $TC(x) = \text{integral of } MC(x) dx = \text{integral of } 2x^1 dx.$

→ $= 1x^2 + C.$

Answer: $\frac{2}{2}x^2 + C \Rightarrow \text{coeff} = 1$

9. Find the antiderivative (indefinite integral): integral of $5x^2 dx$.

$$\int 5x^2 dx$$

→ Power rule for integration: $\text{integral of } x^n dx = x^{(n+1)}/(n+1) + C.$

→ $\text{integral of } \{a\}x^{\{n\}} dx = (\{a\}/\{n+1\})x^{\text{new_exp210}} + C = \{\text{answer_antideriv210}\}x^{\text{new_exp210}} + C.$

Answer: $\frac{5}{3}x^3 + C \Rightarrow \text{coeff} = 5/3$

10. The marginal cost function is $MC(x) = 3x^1$. Find the total cost function $TC(x)$ by finding the antiderivative.
(Leave + C for the constant.)

$$\int 3x^1 dx$$

→ $TC(x) = \text{integral of } MC(x) dx = \text{integral of } 3x^1 dx.$

→ $= 3/2x^2 + C.$

Answer: $\frac{3}{2}x^2 + C \Rightarrow \text{coeff} = 3/2$



11. Find the antiderivative (indefinite integral): integral of $5x^2 dx$.

$$\int 5x^2 dx$$

→ Power rule for integration: integral of $x^n dx = x^{(n+1)}/(n+1) + C$.

→ integral of $\{a\}x^{\{n\}} dx = (\{a\}/\{n+1\})x^{\{new_exp210\}} + C = \{answer_antideriv210\}x^{\{new_exp210\}} + C$.

Answer: $\frac{5}{3}x^3 + C \Rightarrow \text{coeff} = 5/3$

12. The marginal cost function is $MC(x) = 5x^2$. Find the total cost function $TC(x)$ by finding the antiderivative. (Leave + C for the constant.)

$$\int 5x^2 dx$$

→ $TC(x) = \text{integral of } MC(x) dx = \text{integral of } 5x^2 dx$.

→ $= 5/3x^3 + C$.

Answer: $\frac{5}{3}x^3 + C \Rightarrow \text{coeff} = 5/3$

13. Find the antiderivative (indefinite integral): integral of $1x^3 dx$.

$$\int 1x^3 dx$$

→ Power rule for integration: integral of $x^n dx = x^{(n+1)}/(n+1) + C$.

→ integral of $\{a\}x^{\{n\}} dx = (\{a\}/\{n+1\})x^{\{new_exp210\}} + C = \{answer_antideriv210\}x^{\{new_exp210\}} + C$.

Answer: $\frac{1}{4}x^4 + C \Rightarrow \text{coeff} = 1/4$

14. The marginal cost function is $MC(x) = 2x^1$. Find the total cost function $TC(x)$ by finding the antiderivative. (Leave + C for the constant.)

$$\int 2x^1 dx$$

→ $TC(x) = \text{integral of } MC(x) dx = \text{integral of } 2x^1 dx$.

→ $= 1x^2 + C$.

Answer: $\frac{2}{2}x^2 + C \Rightarrow \text{coeff} = 1$

15. Find the antiderivative (indefinite integral): integral of $1x^3 dx$.

$$\int 1x^3 dx$$

→ Power rule for integration: integral of $x^n dx = x^{(n+1)}/(n+1) + C$.

→ integral of $\{a\}x^{\{n\}} dx = (\{a\}/\{n+1\})x^{\{new_exp210\}} + C = \{answer_antideriv210\}x^{\{new_exp210\}} + C$.

Answer: $\frac{1}{4}x^4 + C \Rightarrow \text{coeff} = 1/4$



16. The marginal cost function is $MC(x) = 4x^3$. Find the total cost function $TC(x)$ by finding the antiderivative.
(Leave + C for the constant.)

$$\int 4x^3 dx$$

→ $TC(x) = \text{integral of } MC(x) dx = \text{integral of } 4x^3 dx.$

→ $= 1x^4 + C.$

Answer: $\frac{4}{4}x^4 + C \Rightarrow \text{coeff} = 1$



Definite integrals

17. Evaluate the definite integral: integral from 2 to 2 of x^2 dx.

$$\int_2^2 x^2 dx$$

→ FTC: integral from $\{lo\}$ to $\{hi\}$ of x^n dx = $[x^{n+1}/(n+1)]$ from $\{lo\}$ to $\{hi\}$.

→ = $(\{hi_pow\} - \{lo_pow\}) / \{n+1\} = \{answer_defint210\}$.

Answer: $\frac{8 - 8}{3} = 0$

18. Find the area under the curve $f(x) = x^2$ from $x = 1$ to $x = 3$.

$$\int_1^3 x^2 dx$$

→ Area = integral from $\{lo\}$ to $\{hi\}$ of x^n dx = $[x^{n+1}/(n+1)]$ from $\{lo\}$ to $\{hi\}$.

→ = $(\{hi_pow\} - \{lo_pow\}) / \{n+1\} = \{answer_defint210\}$ square units.

Answer: $\frac{27 - 1}{3} = 26/3$

19. Evaluate the definite integral: integral from 1 to 3 of x^3 dx.

$$\int_1^3 x^3 dx$$

→ FTC: integral from $\{lo\}$ to $\{hi\}$ of x^n dx = $[x^{n+1}/(n+1)]$ from $\{lo\}$ to $\{hi\}$.

→ = $(\{hi_pow\} - \{lo_pow\}) / \{n+1\} = \{answer_defint210\}$.

Answer: $\frac{81 - 1}{4} = 20$

20. Find the area under the curve $f(x) = x^1$ from $x = 1$ to $x = 3$.

$$\int_1^3 x^1 dx$$

→ Area = integral from $\{lo\}$ to $\{hi\}$ of x^n dx = $[x^{n+1}/(n+1)]$ from $\{lo\}$ to $\{hi\}$.

→ = $(\{hi_pow\} - \{lo_pow\}) / \{n+1\} = \{answer_defint210\}$ square units.

Answer: $\frac{9 - 1}{2} = 4$

21. Evaluate the definite integral: integral from 0 to 5 of x^2 dx.

$$\int_0^5 x^2 dx$$

→ FTC: integral from $\{lo\}$ to $\{hi\}$ of x^n dx = $[x^{n+1}/(n+1)]$ from $\{lo\}$ to $\{hi\}$.

→ = $(\{hi_pow\} - \{lo_pow\}) / \{n+1\} = \{answer_defint210\}$.

Answer: $\frac{125 - 0}{3} = 125/3$



22. Find the area under the curve $f(x) = x^1$ from $x = 1$ to $x = 4$.

$$\int_1^4 x^1 dx$$

→ Area = integral from {lo} to {hi} of x^n dx = $[x^{n+1}/(n+1)]$ from {lo} to {hi}.

→ = $(\{hi_pow\} - \{lo_pow\}) / (n+1) = \{answer_defint210\}$ square units.

Answer: $\frac{16 - 1}{2} = 15/2$

23. Evaluate the definite integral: integral from 0 to 5 of x^3 dx.

$$\int_0^5 x^3 dx$$

→ FTC: integral from {lo} to {hi} of x^n dx = $[x^{n+1}/(n+1)]$ from {lo} to {hi}.

→ = $(\{hi_pow\} - \{lo_pow\}) / (n+1) = \{answer_defint210\}$.

Answer: $\frac{625 - 0}{4} = 625/4$

24. Find the area under the curve $f(x) = x^1$ from $x = 1$ to $x = 4$.

$$\int_1^4 x^1 dx$$

→ Area = integral from {lo} to {hi} of x^n dx = $[x^{n+1}/(n+1)]$ from {lo} to {hi}.

→ = $(\{hi_pow\} - \{lo_pow\}) / (n+1) = \{answer_defint210\}$ square units.

Answer: $\frac{16 - 1}{2} = 15/2$

25. Evaluate the definite integral: integral from 1 to 3 of x^2 dx.

$$\int_1^3 x^2 dx$$

→ FTC: integral from {lo} to {hi} of x^n dx = $[x^{n+1}/(n+1)]$ from {lo} to {hi}.

→ = $(\{hi_pow\} - \{lo_pow\}) / (n+1) = \{answer_defint210\}$.

Answer: $\frac{27 - 1}{3} = 26/3$

26. Find the area under the curve $f(x) = x^2$ from $x = 0$ to $x = 4$.

$$\int_0^4 x^2 dx$$

→ Area = integral from {lo} to {hi} of x^n dx = $[x^{n+1}/(n+1)]$ from {lo} to {hi}.

→ = $(\{hi_pow\} - \{lo_pow\}) / (n+1) = \{answer_defint210\}$ square units.

Answer: $\frac{64 - 0}{3} = 64/3$



27. Evaluate the definite integral: integral from 0 to 5 of x^2 dx.

$$\int_0^5 x^2 dx$$

→ FTC: integral from {lo} to {hi} of x^n dx = $[x^{n+1}/(n+1)]$ from {lo} to {hi}.

→ = $(\{hi_pow\} - \{lo_pow\}) / \{n+1\} = \{answer_defint210\}$.

Answer: $\frac{125 - 0}{3} = 125/3$

28. Find the area under the curve $f(x) = x^2$ from $x = 1$ to $x = 3$.

$$\int_1^3 x^2 dx$$

→ Area = integral from {lo} to {hi} of x^n dx = $[x^{n+1}/(n+1)]$ from {lo} to {hi}.

→ = $(\{hi_pow\} - \{lo_pow\}) / (n+1) = \{answer_defint210\}$ square units.

Answer: $\frac{27 - 1}{3} = 26/3$

29. Evaluate the definite integral: integral from 1 to 2 of x^3 dx.

$$\int_1^2 x^3 dx$$

→ FTC: integral from {lo} to {hi} of x^n dx = $[x^{n+1}/(n+1)]$ from {lo} to {hi}.

→ = $(\{hi_pow\} - \{lo_pow\}) / (n+1) = \{answer_defint210\}$.

Answer: $\frac{16 - 1}{4} = 15/4$

30. Find the area under the curve $f(x) = x^1$ from $x = 0$ to $x = 2$.

$$\int_0^2 x^1 dx$$

→ Area = integral from {lo} to {hi} of x^n dx = $[x^{n+1}/(n+1)]$ from {lo} to {hi}.

→ = $(\{hi_pow\} - \{lo_pow\}) / (n+1) = \{answer_defint210\}$ square units.

Answer: $\frac{4 - 0}{2} = 2$

